Supplementary Information: Matter-Dilution-Triggered Recollapse and Conditional Nonsingular Cycles in a Metastable Dark-Energy Model

Online Resource 1 for a manuscript submitted to General Relativity and Gravitation
Leon Sandler, Independent Researcher, Northbrook, Illinois, USA
Corresponding author: sandler.leon@gmail.com

S1. Overview

This supplement documents the numerical methods, the validation tables and the reproduction commands for the paper. Every number quoted in the paper is read from the JSON files produced by the scripts named here, and the manuscript is built only if the checks of Table S8 pass.

S2. The matter-held vacuum and its release

matter_coupling.py defines V_0, A, ρ_{crit} and a_c; dilution_background.py integrates the coupled background (flat, general relativity) from $z = 1100$ with the field at $10^{-60}\,v$, with the absolute tolerance of the field set relative to its seed (an absolute tolerance on the background scale would swamp the seed); turnaround_scan.py runs the grid of Table 1 for the coupled model and the control; dilution_thermal.py the thermal comparison of Section 3.2; causal_controls.py the interventions of Section 3.3; dilution_validation.py the checks of Table S1.

Table S1. Validation of the dilution-trigger code (dilution_validation.py).

Check	Content	Result
D1	constraint	pass
D2	convergence integrators	pass
D3	critical scale factor	pass
D4	LCDM before today	pass
D5	seed independent past	pass
D6	energy relation	pass
D7	turnaround event	pass
D8	thermal saturation	pass
D9	tunneling zero density	pass
D10	effective bounce	pass

Table S2. Release and turnaround over the grid ($q = -0.5$ and 1; the control does not depend on a_c). The full grid, including $q = 0.1$ and 10, is in turnaround_scan.csv.

a_c	μ / H_0	q	model	outcome	turnaround a	residual
1.25	3	-0.5	coupled	transition, no recollapse	–	3e-05
1.25	3	1	coupled	transition and recollapse	1.11e+26	3e-05
1.25	10	-0.5	coupled	transition, no recollapse	–	4e-07
1.25	10	1	coupled	transition and recollapse	1.42e+06	4e-07
1.25	30	-0.5	coupled	transition, no	–	1e-06

				recollapse		
1.25	30	1	coupled	transition and recollapse	123	1e-06
1.25	100	-0.5	coupled	transition, no recollapse	–	3e-07
1.25	100	1	coupled	transition and recollapse	6.74	3e-07
1.25	300	-0.5	coupled	transition, no recollapse	–	2e-06
1.25	300	1	coupled	transition and recollapse	2.98	2e-06
1.5	3	-0.5	coupled	transition, no recollapse	–	8e-06
1.5	3	1	coupled	transition and recollapse	1.15e+26	8e-06
1.5	10	-0.5	coupled	transition, no recollapse	–	1e-06
1.5	10	1	coupled	transition and recollapse	1.6e+06	1e-06
1.5	30	-0.5	coupled	transition, no recollapse	–	4e-06
1.5	30	1	coupled	transition and recollapse	146	4e-06
1.5	100	-0.5	coupled	transition, no recollapse	–	2e-06
1.5	100	1	coupled	transition and recollapse	8.02	2e-06
1.5	300	-0.5	coupled	transition, no recollapse	–	2e-06
1.5	300	1	coupled	transition and recollapse	3.55	2e-06
2	3	-0.5	coupled	transition, no recollapse	–	4e-07
2	3	-0.5	control	transition, no recollapse	–	5e-05
2	3	1	coupled	transition and recollapse	1.88e+26	4e-07
2	3	1	control	transition and recollapse	4.28e+24	5e-05
2	10	-0.5	coupled	transition, no recollapse	–	2e-06
2	10	-0.5	control	transition, no recollapse	–	5e-05
2	10	1	coupled	transition and recollapse	2.09e+06	2e-06
2	10	1	control	transition and recollapse	2.84e+05	5e-05
2	30	-0.5	coupled	transition, no recollapse	–	5e-06
2	30	-0.5	control	transition, no recollapse	–	5e-05
2	30	1	coupled	transition and recollapse	195	5e-06
2	30	1	control	transition and recollapse	34.9	5e-05
2	100	-0.5	coupled	transition, no recollapse	–	1e-06
2	100	-0.5	control	transition, no recollapse	–	4e-05
2	100	1	coupled	transition and recollapse	10.6	1e-06

2	100	1	control	transition and recollapse	2.05	5e-05
2	300	-0.5	coupled	transition, no recollapse	–	3e-06
2	300	-0.5	control	rolled before today	–	1e-05
2	300	1	coupled	transition and recollapse	4.7	4e-06
2	300	1	control	rolled before today	0.949	4e-05
3	3	-0.5	coupled	transition, no recollapse	–	1e-06
3	3	1	coupled	transition and recollapse	2.77e+26	1e-06
3	10	-0.5	coupled	transition, no recollapse	–	2e-06
3	10	1	coupled	transition and recollapse	3.25e+06	2e-06
3	30	-0.5	coupled	transition, no recollapse	–	2e-06
3	30	1	coupled	transition and recollapse	292	2e-06
3	100	-0.5	coupled	transition, no recollapse	–	1e-05
3	100	1	coupled	transition and recollapse	16	1e-05
3	300	-0.5	coupled	transition, no recollapse	–	1e-05
3	300	1	coupled	transition and recollapse	7.04	1e-05
5	3	-0.5	coupled	transition, no recollapse	–	2e-06
5	3	1	coupled	transition and recollapse	1.1e+27	2e-06
5	10	-0.5	coupled	transition, no recollapse	–	6e-06
5	10	1	coupled	transition and recollapse	5.68e+06	6e-06
5	30	-0.5	coupled	transition, no recollapse	–	3e-06
5	30	1	coupled	transition and recollapse	497	3e-06
5	100	-0.5	coupled	transition, no recollapse	–	3e-06
5	100	1	coupled	transition and recollapse	26.7	3e-06
5	300	-0.5	coupled	transition, no recollapse	–	9e-06
5	300	1	coupled	transition and recollapse	11.7	9e-06

S3. Stability of repeated cycles (Section 4.2)

oscillation.py integrates the open negative-vacuum universe with the effective bounce continuously through every bounce, with no reset; oscillation_validation.py repeats the runs with different integrators, tests the continuity at the bounces and the energy-balance law, and maps the phase diagram of Fig. 4b.

Table S3. Scalar energy multiplier per cycle (linear regime) and turnaround radii of a nonlinear run for each integrator setting, with their deviation from the reference.

Integrator	**multiplier**	**deviation**	a_{max} **deviation**
DOP853 1e-11 P/400	10.1634	0.0e+00	0.0e+00
DOP853 1e-12 P/800	10.1742	1.1e-03	2.2e-03
DOP853 1e-9 P/200	8.0140	2.1e-01	2.2e-01
DOP853 1e-11 P/1600	10.1575	5.8e-04	2.2e-04
Radau 1e-10 P/400	10.1713	7.7e-04	2.1e-03
LSODA 1e-11 P/400	9.8939	2.7e-02	4.6e-02

Table S4. Phase diagram of repeated cycles: classification after 100 or more cycles (runs stop when a exceeds 10^{30}).

m	YP	φ_0	**cycles**	**class**
0.01	0.0	1e-08	17	growing (runaway)
0.01	0.0	1e-03	5	growing (runaway)
0.01	0.3	1e-08	20	growing (runaway)
0.01	0.3	1e-03	6	growing (runaway)
0.01	1.0	1e-08	42	growing (runaway)
0.01	1.0	1e-03	7	growing (runaway)
0.01	3.0	1e-08	178	stable
0.01	3.0	1e-03	178	stable
0.01	10.0	1e-08	178	stable
0.01	10.0	1e-03	178	stable
0.03	0.0	1e-08	11	growing (runaway)
0.03	0.0	1e-03	7	growing (runaway)
0.03	0.3	1e-08	13	growing (runaway)
0.03	0.3	1e-03	9	growing (runaway)
0.03	1.0	1e-08	39	growing (runaway)
0.03	1.0	1e-03	32	growing (runaway)
0.03	3.0	1e-08	52	growing (runaway)
0.03	3.0	1e-03	46	growing (runaway)
0.03	10.0	1e-08	178	stable
0.03	10.0	1e-03	178	stable
0.05	0.0	1e-08	20	growing (runaway)
0.05	0.0	1e-03	9	growing (runaway)
0.05	0.3	1e-08	24	growing (runaway)
0.05	0.3	1e-03	13	growing (runaway)
0.05	1.0	1e-08	36	growing (runaway)
0.05	1.0	1e-03	18	growing (runaway)
0.05	3.0	1e-08	178	stable
0.05	3.0	1e-03	178	stable
0.05	10.0	1e-08	178	stable
0.05	10.0	1e-03	178	stable
0.10	0.0	1e-08	28	growing (runaway)
0.10	0.0	1e-03	14	growing (runaway)
0.10	0.3	1e-08	39	growing (runaway)
0.10	0.3	1e-03	19	growing (runaway)
0.10	1.0	1e-08	50	growing (runaway)
0.10	1.0	1e-03	27	growing (runaway)
0.10	3.0	1e-08	178	stable
0.10	3.0	1e-03	198	growing
0.10	10.0	1e-08	178	stable
0.10	10.0	1e-03	178	stable
0.20	0.0	1e-08	27	growing (runaway)
0.20	0.0	1e-03	17	growing (runaway)

0.20	0.3	1e-08	39	growing (runaway)
0.20	0.3	1e-03	32	growing (runaway)
0.20	1.0	1e-08	85	growing (runaway)
0.20	1.0	1e-03	68	growing (runaway)
0.20	3.0	1e-08	179	stable
0.20	3.0	1e-03	180	stable
0.20	10.0	1e-08	178	stable
0.20	10.0	1e-03	178	stable

S4. Continuous cycles, stability, bounce and relics (Sections 4.3–6.2)

continuous_cycles.py integrates the cycle of Section 5.1 as described in Section 6.1; cycle_map.py probes the bounce-to-bounce map; numerics_admissibility.py repeats three baseline cycles with rtol 10^{-10}, an overdamped threshold of 10^{4}, projection at every fourth chunk, and no projection; cycle_scan.py maps the outcome over dark-matter mass and bounce density; contraction_stability.py, bounce_validation.py, compact_relics.py and predictions.py give Sections 4.3–4.5 and 7.1.

Table S5. Continuous cycles (baseline), cycles 1–20 and every tenth cycle thereafter; all cycles in continuous_cycles.csv.

cycle	φ_b / v	Y_{fo} / κ	Y / κ **held**	**peak matter**	**held e-folds**	$\ln G$	$\frac{1}{3}\ln$ **S ratio**
1	0.3783	61.201	1.160	0.996	5.386	7.6616	7.6707
2	0.3753	61.799	0.989	0.995	5.288	7.6662	7.6557
3	0.3758	61.520	1.013	0.995	5.273	7.6619	7.6674
4	0.3754	61.525	1.007	0.995	5.288	7.6647	7.6602
5	0.3756	61.521	1.011	0.995	5.288	7.6628	7.6683
6	0.3754	61.521	1.007	0.995	5.265	7.6646	7.6582
7	0.3757	61.521	1.011	0.995	5.288	7.6622	7.6686
8	0.3754	61.523	1.007	0.995	5.289	7.6646	7.6634
9	0.3755	61.519	1.010	0.995	5.289	7.6640	7.6591
10	0.3757	61.521	1.010	0.995	5.288	7.6625	7.6683
11	0.3754	61.519	1.005	0.995	5.266	7.6646	7.6651
12	0.3755	61.518	1.009	0.995	5.289	7.6640	7.6572
13	0.3758	61.522	1.010	0.995	5.288	7.6621	7.6683
14	0.3754	61.525	1.007	0.995	5.288	7.6646	7.6652
15	0.3755	61.518	1.009	0.995	5.289	7.6640	7.6578
16	0.3757	61.521	1.010	0.995	5.288	7.6626	7.6686
17	0.3754	61.522	1.007	0.995	5.289	7.6645	7.6585
18	0.3756	61.521	1.011	0.995	5.288	7.6627	7.6623
19	0.3756	61.523	1.009	0.995	5.288	7.6628	7.6635
20	0.3756	61.523	1.008	0.995	5.288	7.6632	7.6626
30	0.3758	61.522	1.011	0.995	5.288	7.6620	7.6684
40	0.3757	61.522	1.011	0.995	5.288	7.6623	7.6684
50	0.3758	61.521	1.011	0.995	5.288	7.6621	7.6682
60	0.3754	61.522	1.007	0.995	5.289	7.6645	7.6642
70	0.3755	61.516	1.008	0.995	5.266	7.6641	7.6573
80	0.3756	61.524	1.008	0.995	5.288	7.6634	7.6629
90	0.3755	61.519	1.010	0.995	5.289	7.6639	7.6586
100	0.3754	61.524	1.006	0.995	5.289	–	–

Table S6. Outcome of five continuous cycles per run (cycle_scan.py): held e-folds and Y / κ of cycles 1–5.

Run	**Outcome**	**held e-folds**	Y / κ

$C_T = 0.001$	F1 dark matter not regenerated (bounce too cold)	0.03, 0.01, 0.01, 0.01, 0.01	0.00047, 7.1e-13, 1.4e-25, 2.6e-38, 5e-51
$C_T = 1000$	F1 dark matter not regenerated (bounce too cold)	0.00, 0.00, 0.00, 0.00, 0.00	7e-06, 7.9e-06, 8.3e-06, 8.5e-06, 8.5e-06
$C_T = 1e-06$	F6 wall-clock limit	0.02, 0.01, 0.01, 0.01	3.1e-06, 2.6e-64, 6.9e-132, 1.8e-199
$a_c = 1.5$	working	3.39, 3.55, 3.42, 3.50, 3.45	1.1, 1.1, 1.1, 1.1, 1.1
$a_c = 3$	working	4.59, 5.59, 4.92, 5.22, 5.01	1.1, 1.2, 1.2, 1.2, 1.2
ghost bounce	F6 wall-clock limit		
$m = 1.01$ MeV, $T_b \approx 1.83$ MeV	working	4.03, 4.45, 4.16, 4.34, 4.23	1.2, 1.2, 1.2, 1.2, 1.2
$m = 1.01$ MeV, $T_b \approx 18.3$ MeV	working	4.01, 4.46, 4.16, 4.34, 4.22	1.2, 1.2, 1.2, 1.2, 1.2
$m = 1.01$ MeV, $T_b \approx 183$ MeV	working	4.00, 4.45, 4.16, 4.33, 4.21	1.2, 1.2, 1.2, 1.2, 1.2
$m = 1.01$ MeV, $T_b \approx 1.83e+03$ MeV	working	4.02, 4.47, 4.14, 4.34, 4.22	1.2, 1.2, 1.2, 1.2, 1.2
$m = 10.1$ MeV, $C_T = 0.001$	F1 dark matter not regenerated (bounce too cold)	0.01, 0.01, 0.01, 0.01, 0.01	0.00047, 7.1e-13, 1.4e-25, 2.6e-38, 5e-51
$m = 10.1$ MeV, $C_T = 1000$	F2 vacuum not restored (no held epoch)	0.00, 0.00, 0.00, 0.00, 0.00	1.3, 1.3, 1.3, 1.3, 1.3
$m = 10.1$ MeV, $a_c = 1.5$	working	3.33, 3.55, 3.45, 3.50, 3.44	1.1, 1.1, 1.1, 1.1, 1.1
$m = 10.1$ MeV, $a_c = 3$	working	4.71, 5.45, 4.97, 5.23, 5.12	1.3, 1.2, 1.3, 1.3, 1.3
$m = 10.1$ MeV, $\mu / H_0 = 10$	working	2.23, 3.48, 2.43, 3.09, 2.63	1.4, 1.3, 1.4, 1.3, 1.3
$m = 10.1$ MeV, $\mu / H_0 = 100$	working	4.87, 2.42, 4.83, 2.42, 4.85	1, 1.3, 1, 1.3, 1
$m = 10.1$ MeV, $q = 0.3$	working	2.86, 3.04, 2.99, 2.99, 3.02	1.3, 1.3, 1.3, 1.3, 1.3
$m = 10.1$ MeV, $q = 3$	working	2.76, 2.83, 2.81, 2.84, 2.85	1.3, 1.3, 1.3, 1.3, 1.3
$m = 10.1$ MeV, $T_b \approx 1.83$ MeV	F1 dark matter not regenerated (bounce too cold)	0.00, 0.00, 0.00, 0.00, 0.00	1.8e-06, 1.2e-05, 1.2e-05, 1.2e-05, 1.2e-05
$m = 10.1$ MeV, $T_b \approx 18.3$ MeV	working	3.99, 4.46, 4.16, 4.32, 4.21	1.2, 1.1, 1.2, 1.2, 1.2
$m = 10.1$ MeV, $T_b \approx 183$ MeV	working	4.01, 4.47, 4.14, 4.33, 4.20	1.2, 1.1, 1.2, 1.2, 1.2
$m = 10.1$ MeV, $T_b \approx 1.83e+03$ MeV	working	3.99, 4.46, 4.13, 4.32, 4.22	1.2, 1.1, 1.2, 1.2, 1.2
$m = 101$ MeV, $T_b \approx 1.83$ MeV	F1 dark matter not regenerated (bounce too cold)	0.00, 0.00, 0.00, 0.00, 0.00	1.8e-06, 1.2e-05, 1.2e-05, 1.2e-05, 1.2e-05
$m = 101$ MeV, $T_b \approx 18.3$ MeV	F1 dark matter not regenerated (bounce too cold)	0.00, 0.00, 0.00, 0.00, 0.00	1.8e-06, 1.2e-05, 1.2e-05, 1.2e-05, 1.2e-05
$m = 101$ MeV, $T_b \approx 183$ MeV	working	3.98, 4.45, 4.15, 4.34, 4.22	1.2, 1.1, 1.2, 1.2, 1.2
$m = 101$ MeV, $T_b \approx 1.83e+03$ MeV	working	4.00, 4.45, 4.17, 4.32, 4.20	1.2, 1.1, 1.2, 1.1, 1.2
$m = 1.01e+03$ MeV,	F1 dark matter not regenerated (bounce too cold)	0.00, 0.00, 0.00, 0.00, 0.00	1.8e-06, 1.2e-05, 1.2e-05, 1.2e-05, 1.2e-05

$T_b \approx 1.83$ MeV			
$m = 1.01e+03$ MeV, $T_b \approx 18.3$ MeV	F1 dark matter not regenerated (bounce too cold)	0.00, 0.00, 0.00, 0.00, 0.00	1.8e-06, 1.2e-05, 1.2e-05, 1.2e-05, 1.2e-05
$m = 1.01e+03$ MeV, $T_b \approx 183$ MeV	F1 dark matter not regenerated (bounce too cold)	0.00, 0.00, 0.00, 0.00, 0.00	1.8e-06, 1.2e-05, 1.2e-05, 1.2e-05, 1.2e-05
$m = 1.01e+03$ MeV, $T_b \approx 1.83e+03$ MeV	working	4.02, 4.44, 4.15, 4.31, 4.24	1.2, 1.1, 1.2, 1.1, 1.2
$\mu / H_0 = 10$	F1 dark matter not regenerated (bounce too cold)	0.00, 0.00, 0.00, 0.00, 0.00	1.1e-11, 8.4e-12, 8.4e-12, 8.5e-12, 8.6e-12
$\mu / H_0 = 100$	F1 dark matter not regenerated (bounce too cold)	4.88, 0.00, 0.00, 0.00, 0.00	1, 3.1e-07, 3e-07, 3e-07, 3e-07
$q = 0.3$	F1 dark matter not regenerated (bounce too cold)	0.00, 3.00, 1.94, 4.42, 0.00	2.3e-05, 1.1, 0.13, 1.1, 5.8e-07
$q = 3$	F1 dark matter not regenerated (bounce too cold)	0.00, 0.00, 0.00, 0.00, 0.00	7.5e-06, 1.6e-05, 1.6e-05, 1.6e-05, 1.6e-05
$\Omega_{\sigma,0} = 10^{-10}$	F4 shear-dominated bounce	0.00, 0.00, 0.00, 0.00, 0.00	1.9e-06, 1.2e-05, 1.2e-05, 1.2e-05, 1.2e-05
$\Omega_{\sigma,0} = 10^{-20}$	working	4.00, 4.45, 4.17, 4.33, 4.20	1.2, 1.1, 1.2, 1.2, 1.2

Table S7. Upper part: ekpyrotic phase ($c = 5$), largest controlled initial anisotropy against the ekpyrotic fraction at the turnaround and the exit density. Lower part: percolation temperature of black-hole populations in the contraction.

f_0 / **population**	ρ_{end} / ρ_c	s_{max} / **T**
1e-20	1e-08	4.6×10^{-21}
1e-20	0.0001	4.6×10^{-21}
1e-20	0.01	4.6×10^{-21}
1e-12	1e-08	2.1×10^{-16}
1e-12	0.0001	3.5×10^{-13}
1e-12	0.01	9.5×10^{-13}
1e-08	1e-08	2.1×10^{-8}
1e-08	0.0001	3.6×10^{-8}
stellar 10 Msun		2.29e+08 MeV
SMBH 1e8 Msun		1.16e+04 MeV
SMBH 1e9 Msun		2.5e+03 MeV
SMBH 1e10 Msun		539 MeV

S5. Executable verification

Table S8. Checks of this paper (verify.py) and their key figures.

No.	What is checked	Outcome	Key figures
V27	matter-held vacuum: held today, release, recollapse; control fails	pass	validation_checks = 10; scan_runs = 200; max_residual = 4.73e-05
V28	prescribed-step hot cycles (earlier version) and their controls	pass	runs = 8; max_spread_between_cycles = 0.000594; controls_fail = True
V29	continuous cycles: 100 cycles	pass	cycles = 100;

	with dark matter regenerated, matter era, held epoch; growth = entropy^$1/3$; cold bounce fails		max_lnG_minus_entropy = 0.0105; max_residual = 5.29e-06
V30	causal test: frozen matter never recollapses; diluting model recollapses for every $q>0$	pass	runs = 100; q_positive = 75; dilution_causes_recollapse = 75
V31	shear scaling and ekpyrotic gain; bounce invariants and continuity; black-hole volume fraction	pass	s_max_no_ek = 1.66e-27; s_max_ek = [1.663678238554056e-13, 3.508859797842641e-13, 1.0233160024924844e-11]; R_over_rho_c = 4
V32	attracting fixed point; closed-form held epoch; numerical devices; independent solver	pass	fixed_point = 0.376; slope = -0.136; held_rms = 0.0202
V33	phase structure (spinodal release) and coherence of the release	pass	seed_over_fluctuation = 10.8; eps_hat_min = 4.42e-64

S6. Sensitivity of the black-hole bound (Section 4.5)

relic_sensitivity.py evolves the black-hole population of compact_relics.py through the contraction of the baseline cycle with one prescription changed at a time, and extrapolates the fraction of space inside horizons from the bounce with $f_V \propto T^3$. The bound is given for two thresholds: $f_V = 1$ (horizon volumes added) and $f_V = 0.34$ (continuum percolation of randomly placed, overlapping spheres). The last column of the lower block gives the joint viability of Section 6.3 with the window closed at that estimate.

Table S9. Black-hole bound on the bounce temperature under changes of the population and accretion prescriptions (baseline bounce 172 MeV).

Variant	f_V **at the bounce**	T_{perc}, $f_V = 1$ **(GeV)**	T_{perc}, **overlap (GeV)**
baseline (E3)	4.91×10^{-4}	2.17	1.52
stellar abundance x 1/3	4.91×10^{-4}	2.17	1.52
stellar abundance x 3	4.91×10^{-4}	2.17	1.52
supermassive abundance x 1/3	1.64×10^{-4}	3.14	2.19
supermassive abundance x 3	1.48×10^{-3}	1.50	1.05
supermassive cut-off 1e9	1.84×10^{-4}	3.02	2.10
supermassive cut-off 2e10	4.71×10^{-3}	1.02	0.71
supermassive center 1e9	1.80×10^{-3}	1.41	0.98
accretion x 0.1	2.09×10^{-4}	2.89	2.02
accretion x 2	1.00×10^{-3}	1.71	1.20
accretion x 3	3.11×10^{-3}	1.18	0.82
accretion x 10	2.18×10^{2}	0.03	0.02
no accretion, no mergers (E2)	1.87×10^{-4}	3.00	2.09
no mergers	4.91×10^{-4}	2.17	1.52

Table S10. Joint viability of the held branch (viability_mc.py, same draws) with the bounce window closed at each estimate of the bound.

Estimate	T_{perc} (GeV)	bounce window	joint
baseline estimate	2.17	66.1%	6.76%
overlap threshold	1.52	62.1%	5.70%
lowest population variant, overlap	0.71	54.0%	3.91%
region overdense by 100, overlap	0.33	45.4%	2.48%
accretion x 10, overlap	0.02	14.9%	0.20%

S7. Independent solver (Section 6.1)

independent_solver.py repeats the baseline (three cycles) with continuous_cycles.run(gravity='constrained'): away from the turning points the Hubble rate is computed from the Friedmann constraint, so no projection is applied; near a turnaround or bounce (where H^2 falls below ε^2 times the natural rate squared, ε the bridge parameter, or the density exceeds $0.4\,\rho_c$) it is evolved with the Raychaudhuri equation. Table S11 lists every cycle of every setting.

Table S11. Cycles of the projected and constrained solvers (cycle 0 starts today).

Setting	cycle	$\ln a_{max}$	t_{cycle}	held e-folds	Y/κ held	$\ln G_n$
projected (baseline)	0	3.12805	5.2980	2.8434	–	7.78404
projected (baseline)	1	10.91209	8.6900	5.3855	1.1622	7.66380
projected (baseline)	2	18.57589	8.5839	5.2888	0.9898	7.66554
projected (baseline)	3	26.24143	8.5808	5.2733	1.0119	–
constrained, bridge 0.05	0	3.12806	5.2980	2.8434	–	7.78792
constrained, bridge 0.05	1	10.91598	8.6946	5.3855	1.1645	7.66185
constrained, bridge 0.05	2	18.57783	8.5816	5.2889	0.9886	7.66628
constrained, bridge 0.05	3	26.24411	8.5816	5.2885	1.0119	–
constrained, bridge 0.02	0	3.12807	5.2980	2.8434	–	7.78794
constrained, bridge 0.02	1	10.91601	8.6946	5.3857	1.1641	7.66176
constrained, bridge 0.02	2	18.57777	8.5815	5.2768	0.9921	7.66648
constrained, bridge 0.02	3	26.24425	8.5819	5.2890	1.0105	–
constrained, bridge 0.05, rtol 1e-10	0	3.12806	5.2980	2.8385	–	7.78786
constrained, bridge 0.05, rtol 1e-10	1	10.91592	8.6945	5.3606	1.1687	7.66181
constrained, bridge 0.05, rtol 1e-10	2	18.57773	8.5816	5.2858	0.9905	7.66627

constrained, bridge 0.05, rtol 1e-10	3	26.24400	8.5816	5.2883	1.0077	–

S8. Reproduction

Environment of the published run: Python 3.11.9, NumPy 2.4.6, SciPy 1.17.1 (Windows). Requirements: NumPy, SciPy, Matplotlib, python-docx. From the code directory: matter_coupling.py and dilution_background.py (the background), turnaround_scan.py, dilution_thermal.py, causal_controls.py, dilution_validation.py, oscillation.py and oscillation_validation.py, continuous_cycles.py (calibrate, then run; several hours), cycle_map.py, numerics_admissibility.py, independent_solver.py, cycle_scan.py, contraction_stability.py, bounce_validation.py, compact_relics.py, relic_sensitivity.py, predictions.py, vacuum_phase_map.py (coherence), viability_mc.py, the figure scripts, verify.py, then paper1_build.py and paper1_supplement.py. Code: https://github.com/sandlerleon/late-time-vacuum-decay, archived at https://doi.org/10.5281/zenodo.23849853.